

Benchmark from the Kochen–Specker and the Ramsey Problems

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I. INTRODUCTIONS

We provide combinatorial instances from two hard problems in quantum foundations and extremal graph theory. The instances contain a large amount of symmetries. These instances (Table I) can be solved efficiently using the SAT+CAS paradigm that breaks symmetries in the search space, but are challenging and interesting for SAT solvers [1]–[3].

II. THE KOCHEN–SPECKER PROBLEM

The Kochen–Specker problem is an open problem in quantum foundations, which can be translated a searching problem on simple graphs. KS_1 represents graphs that satisfy the following properties:

- The graph does not contain the 4-cycle graph C_4 as a subgraph.
- Each vertex of the graph has a minimum degree 3.
- Every vertex is part of a 3-cycle triangle graph C_3 .
- The graph is non-010-colorable, meaning that there is no $\{0, 1\}$ -coloring of the vertices such that both conditions are satisfied:
 - 1) No two adjacent vertices are colored 1.
 - 2) For each triangle, the vertices are not all colored 0.

KS_2 represents graphs that satisfy the following properties:

- The graph does not contain the 4-cycle graph C_4 as a subgraph.
- Each vertex of the graph has a neighbor.
- Every edge is part of a 3-cycle triangle graph C_3 .
- The graph is non-010-colorable.

III. THE RAMSEY PROBLEM

The Ramsey Theorem states that for every $p, q \in \mathbb{Z}$, there exists an $n \in \mathbb{Z}$ such that any red / blue coloring of the edges of the complete graph of order n , denoted K_n , contains a blue monochromatic p -clique or a red monochromatic q -clique. A (p, q) -graph is a graph without a p -clique and without an

Instance Name	Satisfiability
$KS_1(16, 0.3)$	UNSAT
$KS_1(16, 0.4)$	UNSAT
$KS_1(16, 0.5)$	UNSAT
$KS_1(17, 0.4)$	SAT
$KS_1(17, 0.5)$	SAT
$KS_2(17, 0.3)$	UNSAT
$KS_2(17, 0.4)$	UNSAT
$KS_2(17, 0.5)$	UNSAT
$KS_2(18, 0.3)$	UNSAT
$KS_2(18, 0.4)$	UNSAT
$KS_2(18, 0.5)$	UNSAT
$R(25, 4, 5; 12, 12)$	UNSAT

TABLE I
BENCHMARK LISTS

independent set of size q . $(n, p, q; d_1, d_2)$ -graphs are (p, q) -graphs of order n where the degree d of each vertex satisfies $d_1 \leq d \leq d_2$.

IV. BENCHMARKS

We provide instances that can be solved by MiniSat in anywhere from a minute to an hour. We denote instances from the Kochen–Specker problem as $KS_1(n, c)$ and $KS_2(n, c)$, where KS_1, KS_2 represents two different definitions of the Kochen–Specker problem, n is the order of the graph and c is the maximum percentage of color-1 vertices that the encoding blocks. Specifically, only 010-colorings with less than or equal to $\lceil nc \rceil$ color-1 vertices are blocked. We denote instance from the Ramsey problem as $R(n, p, q; d_1, d_2)$ as described in the previous section.

REFERENCES

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